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Revision

Revision index	Entry date	Change description	Person responsible, name, signature, date	CR Number	Remark	Page
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-	22.07.2021	New drawing	M.Wetzel	--	-	0
a	06.10.2021	changed connection of relays	M.Wetzel	CR020522	CN029673	21,29,30,31,32,33

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Revision	Date	Name				Customer Description	Project Description
			Creator	M.Wetzel	22.07.2021		circuit-diagram control-panel BR2000-ECU9 without ctrl.-system
			Editor	M.Wetzel	21.10.2021		
			Appr	F.Jäger	23.07.2021		
			Norm	F.Jäger	23.07.2021		

A Rolls-Royce solution

revisions-overview

Project template with identifier structure to IEC 81346

Drawing No. XZ54634000005 (ZNG00040703)

Material No. XZ54634000029 (ZNG00040704)

Order No --

Department TDEC

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A Rolls-Royce
solution

project-description
circuit-diagram control-panel BR2000-ECU9 without ctrl.-system

order number
--

material number
X54634400029 (ZNG00040704)

drawing-number
XZ54634000005 (ZNG00040703)

serial number

type-of-system
BR2000-ECU9

0 2

			Creator	M.Wetzel	22.07.2021		circuit-diagram control-panel BR2000-ECU9 without ctrl.-system	A Rolls-Royce solution	Cover sheet	Drawing No. XZ54634000005 (ZNG00040703)	Order No --	EAA
			Editor	M.Wetzel	21.10.2021							
			Appr	F.Jäger	23.07.2021					Material No. X54634400029 (ZNG00040704)	Department TDEC	XZ54634000005 (ZNG00040703) Page 1 Pages 69
Revision	Date	Name	Norm	F.Jäger	23.07.2021	Customer Description	Project Description		Project template with identifier structure to IEC 81346			

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		EDB		5	Technical data	06.10.2021	M.Wetzel
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			Editor	M.Wetzel	21.10.2021					Material No. XS4634400029 (ZNG00040704)	Department TDEC	
			Appr	F.Jäger	23.07.2021				Project template with identifier structure to IEC 81346			XZ54634000005 (ZNG00040703) Page 3 Pages 69
			Norm	F.Jäger	23.07.2021							

TECHNICAL DATA AND CONSTRUCTION DETAILS

IEC 61355

Full designation	Structure description
mounting-location	
+BSF	Baseframe
+ENG	Engine
+GEN	Generator incl. terminal box
+MIP	MTU Genset interface panel
+MMC	MTU Genset control
+GCB	Generator circuit breaker
+MCB	Mains circuit breaker
+FAN	Fan control
+TNK	Fuel tank
+EXT	Devices and enclosures outside any other enclosure, not delivered by MTU
+CPL	Other control panel enclosure
+CUS	Customer page
+DOK	documents

Full designation	Structure description
document-type	
&EAA	Electrical cover sheet
&EAB	Electrical table of contents
&EDB	Description
&EFS	Wiring diagram
&ETL	Layout
&EMA	Electrical terminal diagram
&EMB	Electrical cable plan
&EPC	Bill of material

Full designation	Structure description
system	
=EKW	Electrical power
=CON	Control voltages
=EES	Engine emergency stop
=COM	Communication
=STR	Engine starting
=MCC	Mixture cooling circuit
=ECC	Engine/jacket water circuit
=HWC	Heating water circuit
=FUL	Fuel supply

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TECHNICAL DATA AND CONSTRUCTION DETAILS

line and color definition for potential tracing

(Colors are only visible if you use colored print out)

Supply, Line and measure voltage	L1, L2, L3 (120V, 230V, 400V ,690V, ...)	L
Supply, Line and measure voltage	N	N
Protective earth	Ground	PE
Control voltage + 24VDC, 12VDC L+	L+	L+
Control voltage - 0VDC L-	L-	L-
Shield	SH	shield
Starter battery cabeling +24V	L-	L+Batt
Starter battery cabeling -24V	L-	L-Batt

line and color definition for potential tracing

(Colors are only visible if you use colored print out)

Control voltage 24VAC 24VAC L		24VAC-L
Control voltage 24VAC 24VAC N		24VAC-N
External source voltage	....	ext
Analog signals	QSOLL+, QSOLL-	analog
Current transformer	L1-l-Geno, L1-k-Geno	L1-k-Geno
Emergency stop	L+	NA
All other connections EPLAN standard		

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TECHNICAL DATA AND CONSTRUCTION DETAILS

Wire colors within enclosure

According
IEC EN 60204-1

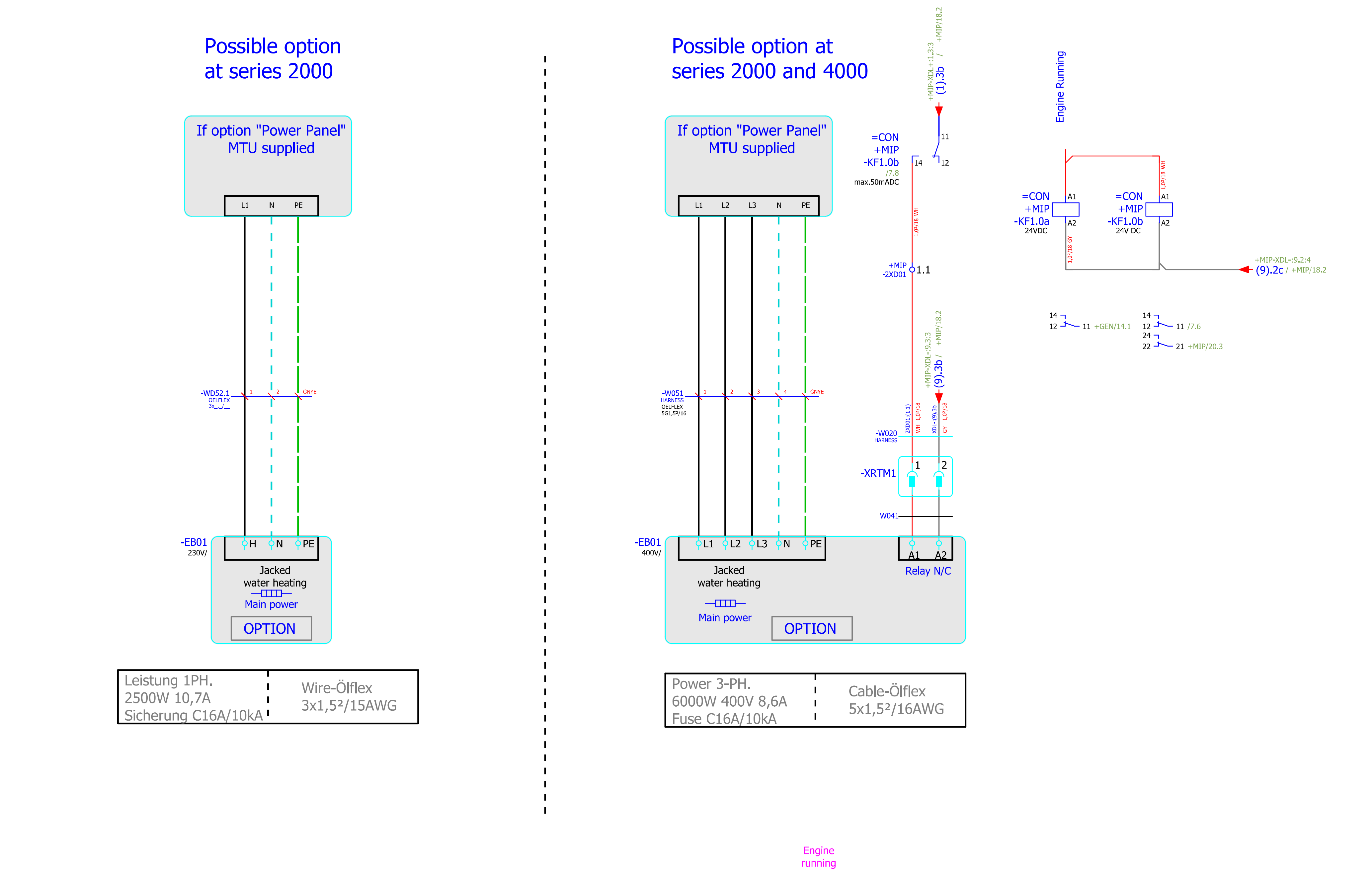
Rated voltage	Above 48VAC	black
Neutral wire	N	light blue
Protective earth	PE	green/yellow
Control voltage	24-240V AC	red
	0V AC	red/white
	24-110V DC	dark blue
	0V DC	grey
PLC-Controller		white
Measuring technique		grey
wires before Mainswitch		orange

Main power before main switch	L1 - L2 - L3
Main power main switch on	1L1 - 1L2 - 1L3
Neutral wire	N
Protective conductor	PE

Power supply	400VAC	-XD01
Power supply	230VAC	-XD02
Auxiliary drives		-XD03
Emergency stop		-XD06
Module signals digital		-XD10
Module signals analog		-XD11
Supply DC	24VDC	-XDL+
Supply DC	0VDC	-XDL-
Signals external DC		-XD100.x
Harness W010-ECU7 / W011-ECU9 Digital		-1XD01
Harness W010-ECU7 / W011-ECU9 Analog		-1XD02
Harness W020 Engine		-2XD01
Harness W030 Generator signal		-3XD01/-3XD02
Harness W031 Generator PT100		-3XD03
Option		
Harness W050 Power panel		-5XD01
Control cable WD06.1 Generator circuit breaker		-6XD01
Control cable WD07.1 Mains circuit breaker		-7XD01
Control cable WD08.1 Fan control enclosure		-8XD01
Control cable WD09.1 Fuel control		-9XD01
CAN bus		-XF01
MOD bus		-XF02

Tightening torque

Power relay	75A	M3,5x5 M5x8	1,1-1,2 Nm 3,2-3,5 Nm
Starter	Prestolite 9KW	M5 M12	2,0-3,0 Nm 27-33 Nm
Starter	Prestolite 15KW	M5 M10	1,5-1,7 Nm 9-10 Nm
Starter	Delco 9KW	Housing 10-32 INCH 1/2-13 INCH	18-23 Nm 1,8-3,4 Nm 27-34 Nm
Battery disconnect switches		M10 M12	15-20 Nm 18-22 Nm
Battery terminals	DIN 72331	Cable M6 M8	5-6 Nm Class 4,6 / 3,5 Nm Class 4,6 / 8,4 Nm
Grounding		M8 M10 M12 M14	Class 8,8 / 28 Nm Class 8,8 / 54 Nm Class 8,8 / 93 Nm Class 8,8 / 148 Nm



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Revision	Date	Name					
			Creator	M.Wetzel	22.07.2021		
			Editor	M.Wetzel	21.10.2021		
			Appr	F.Jäger	23.07.2021		
			Norm	F.Jäger	23.07.2021	Customer Description	Project Description

circuit-diagram control-panel BR2000-ECU9 without ctrl.-system

A Rolls-Royce solution

Coolant water preheating

Project template with identifier structure to IEC 81346

Drawing No. XZ54634000005 (ZNG00040703)

Order No --

Material No. XZ54634000005 (ZNG00040703)

Department

BSF EFS HWC

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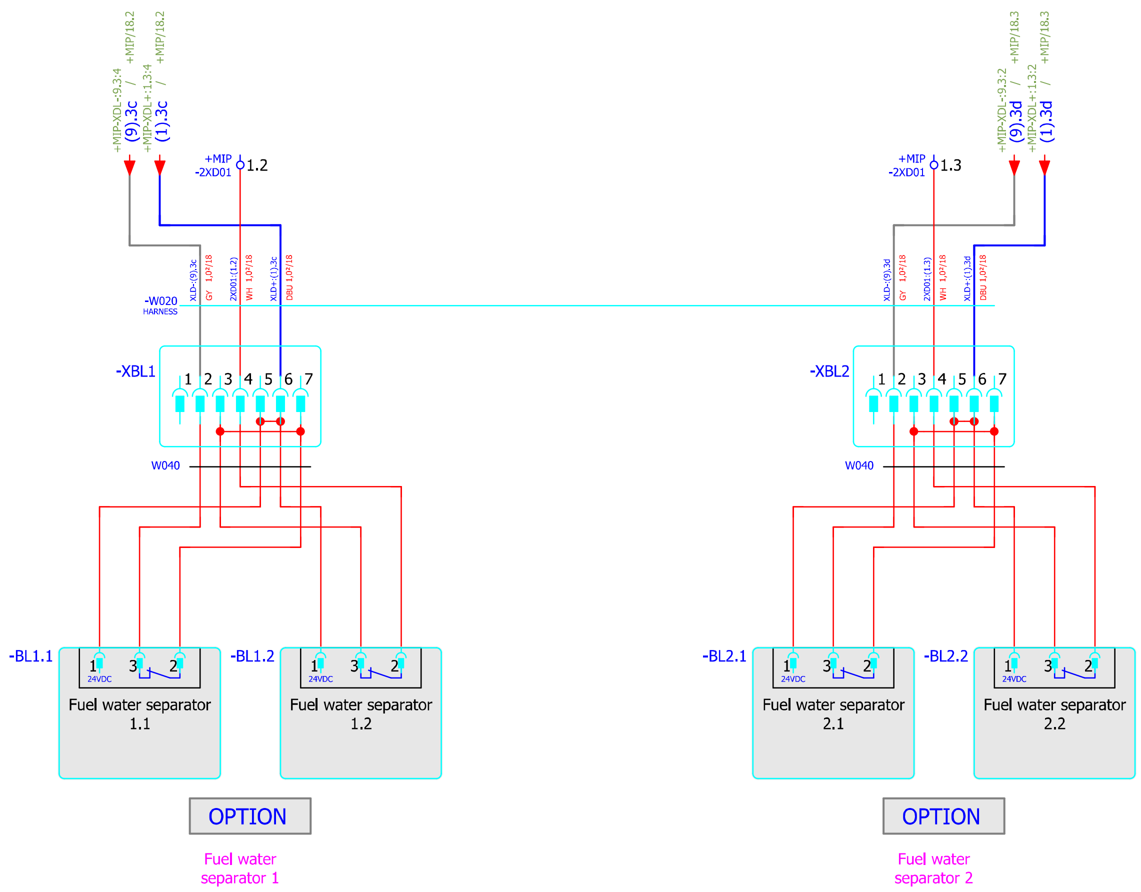

Revision	Date	Name				Customer Description	Project Description
			Creator	M.Wetzel	22.07.2021		circuit-diagram control-panel BR2000-ECU9 without ctrl.-system
			Editor	M.Wetzel	21.10.2021		
			Appr	F.Jäger	23.07.2021		
			Norm	F.Jäger	23.07.2021		

mtu A Rolls-Royce solution

Fuel water separator 1-4

Project template with identifier structure to IEC 81346

Drawing No. XZ54634000005 (ZNG00040703)

Material No. XZ54634000029 (ZNG00040704)

Order No --

Department TDEC

BSF EFS FUL

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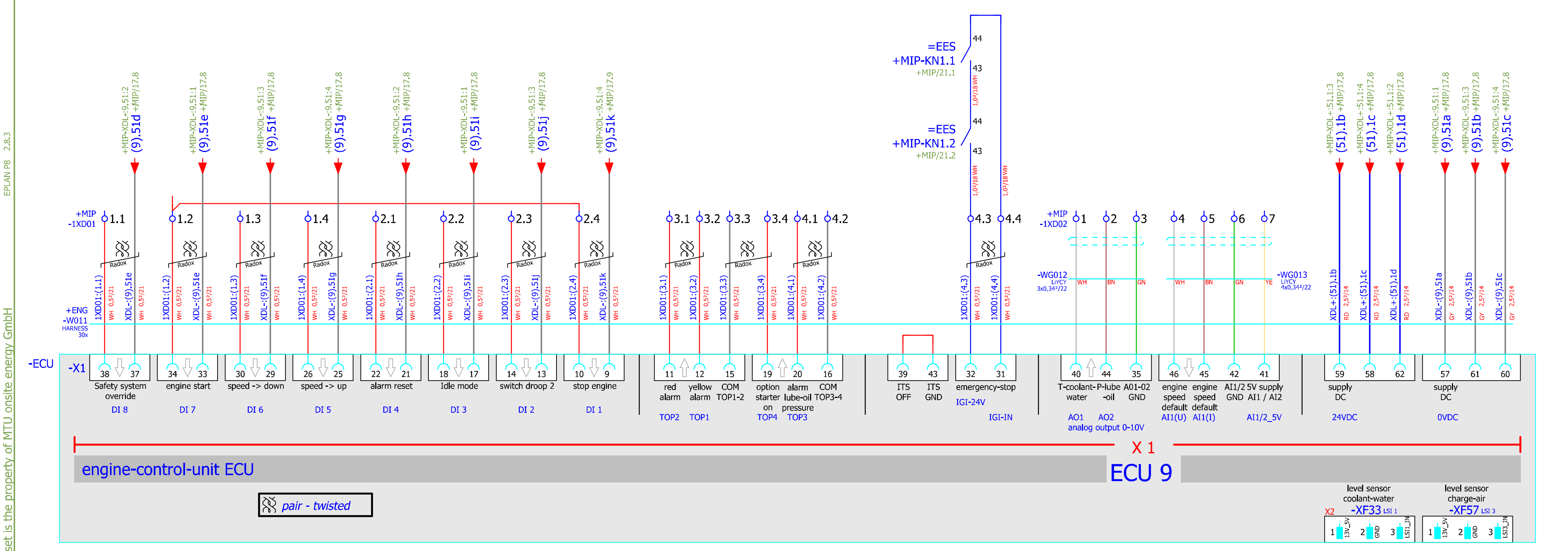

+BSF/8

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Revision	Date	Name				Customer Description	Project Description
			Creator	M.Wetzel	22.07.2021		circuit-diagram control-panel BR2000-ECU9 without ctrl.-system
			Editor	M.Wetzel	21.10.2021		
			Appr	F.Jäger	23.07.2021		
			Norm	F.Jäger	23.07.2021		

mtu — A Rolls-Royce solution

Engine-Control-Unit ECU9
Plug -X1

Project template with identifier structure to IEC 81346

Drawing No. XZ54634000005 (ZNG00040703)
Material No. XZ54634000029 (ZNG00040704)
Order No --
Department TOEE

ENG EFS CON

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